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Assignment-3 Parallel Scientific Computing

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1) sequential part $\sigma(n) = an+b$; $a=10$, $b=1000$

parallelizable part $\phi(n) = cn^3 + dn^2 + en$; $c = 10^{-3}$, $d = 0.2$, $e = 100$

and $n = 1000$

communication overhead @ $\rightarrow$ $\log_{10} n$ locations and each costs

$[1000(\log_{10} p) + (n/10)]$; $p \rightarrow$ number of processes, $p = 100$

(a) speedup $\psi(n,p) = \dfrac{\sigma(n) + \phi(n)}{\sigma(n) + \frac{\phi(n)}{p} + k(n,p)} = \dfrac{\text{~~sequential execution~~}}{\text{parallel execution}}$

$$= \frac{\text{serial} + \text{parallel}}{\text{serial} + \frac{\text{parallel}}{p} + \text{communication}}$$

$$= \frac{an+b + cn^3 + dn + en}{an+b + \frac{cn^3 + dn + en}{p} + \log_{10} n \times [1000(\log_{10} p) + (n/10)]}$$

$$= \frac{10(1000) + 1000 + (10^{-3})(1000)^3 + 0.2(1000)^2 + 100(1000)}{10(1000) + 1000 + \frac{(10^{-3})(1000)^3 + 0.2(1000)^2 + 100(1000)}{100} + \log_{10} 1000 \left\{1000\left[(\log_{10} 100)\right] + \frac{1000}{10}\right\}}$$

$\div$ by 1000

$$= \frac{10 + 1 + 1000 + 200 + 100}{10 + 1 + \frac{1000 + 200 + 100}{100} + \log_{10} 1000\left[\log_{10} 100 + \frac{1}{10}\right]} = \frac{1311.2}{\cancel{28.302}\ 30.3} = \cancel{39.2622}$$

$$= 43.27$$

(b) Efficiency ε of the program,

$$\varepsilon = \frac{\psi(n,p)}{p} = \frac{39.2622}{100} = \underline{\underline{0.39}} \approx \underline{\underline{39\%}}$$

(c) Sequential fraction of the program,

$$f = \frac{\sigma(n)}{\sigma(n) + \frac{\varphi(n)}{p}} = \frac{10(1000) + 1000}{10(1000) + 1000 + (10^{-3})(1000)^3 + 0.2(1000)^2 + 100(1000)}$$

÷ by 1000

$$= \frac{10 + 1}{\frac{10 + 1 + 1000 + 200 + 100}{200}} = \frac{11}{1311.2} = \cancel{9.899 \times 10^{-3}}\ 8.39 \times 10^{-3}$$

$$= \cancel{0.98\%}\ \underline{\underline{0.84\%}}$$

(d) Amdahl speedup for 100 processes,

$$S(n,p) = \frac{1}{f + \frac{(1-f)}{p}} = \frac{1}{8.4 \times 10^{-3} + \frac{1 - 8.4 \times 10^{-3}}{100}}$$

$$= \frac{1}{8.4 \times 10^{-3} + 9.9 \times 10^{-3}} = \frac{1}{18.3 \times 10^{-3}} = \frac{1000}{18.3} \approx \cancel{50.76}$$

$$\approx \underline{\underline{54.64}}$$

(e) Maximum Speedup, (∞ processors)

$$S(n, \infty) = \frac{1}{f} = \frac{1}{8.4 \times 10^{-3}} = \frac{1000}{8.4} \approx \cancel{102.04} \approx \underline{\underline{119.05}}$$

Assignment 3 Kanak Agarwal

ID5130

2) Recursive Doubling Method.

$$3x_1 - x_2 = 2$$
$$-x_1 + 3x_2 - x_3 = 1$$
$$-x_2 + 3x_3 - x_4 = 1$$
$$-x_3 + 3x_4 = 2$$

Assembly,

$$\left[\begin{array}{cccc|c} 3 & -1 & 0 & 0 & 2 \\ -1 & 3 & -1 & 0 & 1 \\ 0 & -1 & 3 & -1 & 1 \\ 0 & 0 & -1 & 3 & 2 \end{array}\right]$$

Row-echelon form

first make lower half 0,

$R_2 \leftarrow R_2 + \frac{1}{3}R_1$ and $R_4 \leftarrow R_4 + \frac{1}{3}R_3$

$$\left[\begin{array}{cccc|c} 3 & -1 & 0 & 0 & 2 \\ 0 & 8/3 & -1 & 0 & 5/3 \\ 0 & -1 & 3 & -1 & 1 \\ 0 & -1/3 & 0 & 8/3 & 7/3 \end{array}\right]$$

$R_3 \leftarrow R_3 + \frac{3}{8}R_2$ and $R_4 \leftarrow R_4 + \frac{1}{8}R_2$

$$\left[\begin{array}{cccc|c} 3 & -1 & 0 & 0 & 2 \\ 0 & 8/3 & -1 & 0 & 5/3 \\ 0 & 0 & 21/8 & -1 & 13/8 \\ 0 & 0 & -1/8 & 8/3 & 19/8 \end{array}\right]$$

now $R_4 \leftarrow R_4 + \frac{1}{21} R_3$

$$\left[\begin{array}{cccc|c} 3 & -1 & 0 & 0 & 2 \\ 0 & 8/3 & -1 & 0 & 5/3 \\ 0 & 0 & 21/8 & -1 & 13/8 \\ 0 & 0 & 0 & 165/63 & 165/63 \end{array}\right]$$

Now make upper half 0,

$R_3 \leftarrow R_3 + \frac{8}{165} R_4$; $R_2 \leftarrow R_2 + \frac{8}{21} R_3$

$R_1 \leftarrow R_1 + \frac{1}{3} R_2$;

$$\left[\begin{array}{cccc|c} 3 & 0 & 0 & 0 & 3 \\ 0 & 8/3 & 0 & 0 & 8/3 \\ 0 & 0 & 21/8 & 0 & 21/8 \\ 0 & 0 & 0 & \frac{165}{63} & 165/63 \end{array}\right]$$

$\therefore$ $\boxed{x_1 = x_2 = x_3 = x_4 = 1.}$

Parallel Scientific Computing - ID5130

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1 Cholesky Decomposition

Cholesky decomposition is a numerical method used to factor a positive-definite matrix into the product of a lower triangular matrix and its conjugate transpose. The decomposition is expressed as,

$$A = LL^T$$

where A is the original positive-definite matrix, L is the lower triangular matrix, and L^T is the conjugate transpose of L. The computation of the Cholesky decomposition is as follows,

$$L_{ij} = \begin{cases} i = j \quad \rightarrow \quad L_{ii} = \sqrt{A_{ii} - \sum_{k=1}^{i-1} L_{ik}^2} \\ i > j \quad \rightarrow \quad L_{ij} = \frac{1}{L_{jj}} \left(A_{ij} - \sum_{k=1}^{j-1} L_{ik} L_{jk} \right) \end{cases}$$

2 Discussion

OpenACC performs worse than the serial implementation for all cases since it is communication dominated and even for the largest matrix size, the usage of the GPU is not justified. The error between the serial and parallel implementations is negligible, which indicates that the parallel implementation is correctly computing the Cholesky decomposition. Further, the reconstruction error of the original matrix from the decomposed matrices is also negligible, confirming the accuracy of the decomposition. The results suggest that for small to medium-sized matrices, the overhead of parallelization may outweigh the benefits, and it may be more efficient to use a serial implementation. However, for larger matrices, the performance of the parallel implementation may improve, but it is crucial to consider the trade-off between computation and communication overhead when deciding whether to use parallelization.

3 Results

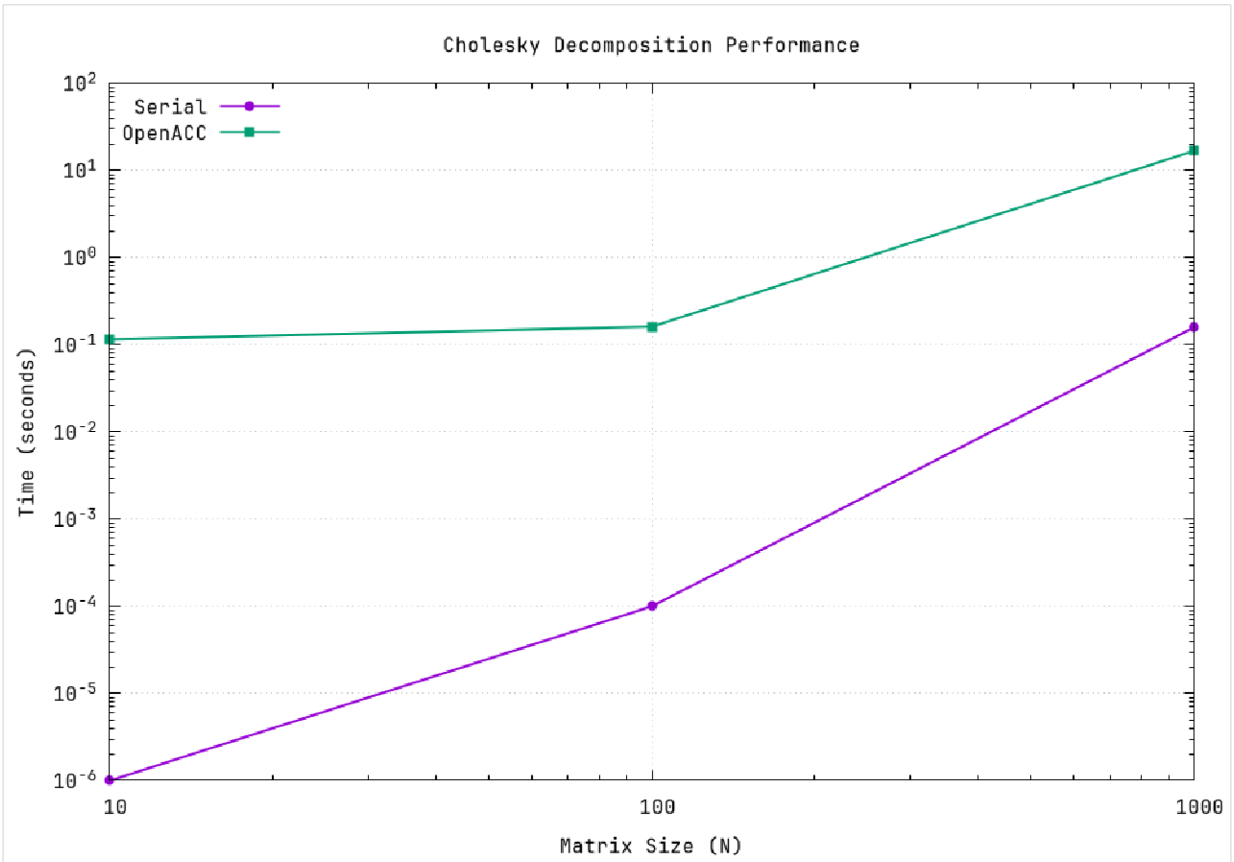

Figure 1: Cholesky Decomposition Timing

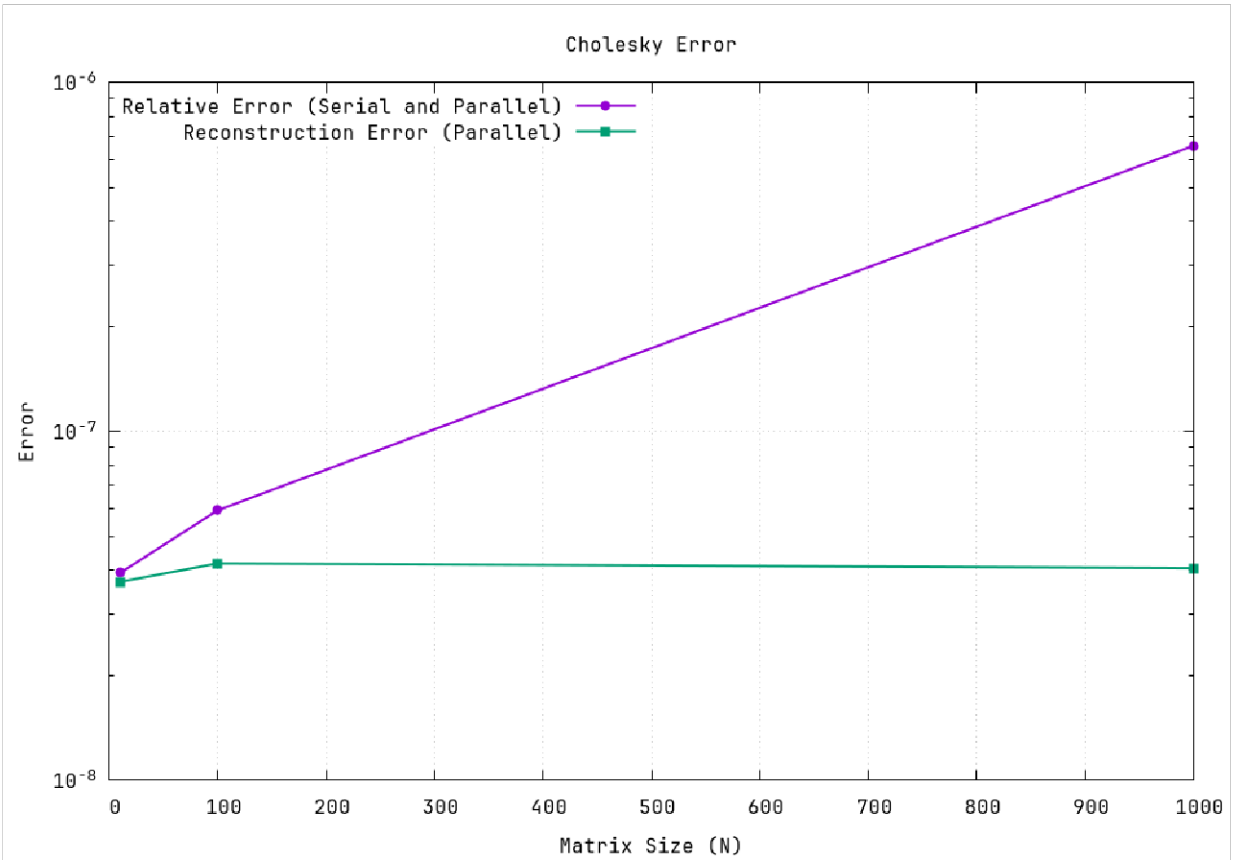

Figure 2: Cholesky Decomposition Errors